Predicting Destruction: Demographic and Spatial Determinants of Violence in the 2002 Gujarat Riots.

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Abstract

This paper examines the relationship that demographic determinants from communities in Gujarat have for intensity of riots after an exogenous event. In February 2002, a train bombing in Godhra Gujarat caused large riots lasting months throughout the province, with many in current literature finding Gujarati Media and religious composition as potential factors for violence. Using demographic data from the 2001 India Census along with data on the destruction of Muslim religious buildings, I construct a model that exploits demographic variation across district to determine if demographic composition of a district can determine riot intensity. Findings showed that there were no statistically significant effects that particular data has on riot intensity. However, the signs of language usage and religious data follow with literature, albeit without statistical significance. These results suggest that there should be a reconsideration in the literature for the importance of language determinants as a cause of riot intensity.

3rd Draft Feedback

For my 3rd draft, I want to have feedback on flow of ideas in the paper. In my last drafts, I was suggested to explain the rationale for my dependent variables. I am less concerned with my empirical model now, I am more concerned with how I foreground everything in the paper and if it makes sense to someone without a South Asian Studies background. I am also figuring out citations, so that is why there is [CITE] in my paper instead of actual in-text citations. There is no appendix right now that will be coming later.

1 Introduction

The 2002 Gujarat riots offers analysis on how composition of a community can be determinant for the destructiveness of inter-communal riots. In different districts of Gujarat, there was variation in intensity of riots. While some cities like Ahmadabad faced months of arson and vandalism, cities like Bhuj didn't receive that same impact. My question is: What impact does social and physical composition of a district have on religious inter-communal violence, given an catalyzing incident. I start with reviewing relevant literature about inter-communal riots with political and economic motivations for the 2002 riots in Gujarat. Then, I explain the process of gathering and choosing data for this analysis. An empirical model based on multiple linear regression using OLS is created which shows inconclusive results for the impact of language of having a causal effect on riots.

1.1 Motivation

I hope to explore this topic because there is a gap in the literature. Moreover, the topic of the 2002 riots provides many insights into how exogenous events that cause violence are impacted given different factors in a community. After Partition, Muslim communities in India have faced greater backlash from the majority Hindu population. I am particularly inspired by my former professor Dr. Zara Choudhary andd her accounts of living in Ahmedebad during the

2002 riots. I would highly recommended reading her novel [] which account the destruction and loss faced from inter-communal riots. More than that, there is a current effort to erase the memories that persisted from the days after February 27th 2002. Current Prime Minister of India, Narendra Modi, led Gujarat during the time of riots and many state he could have done more to prevent the destruction of Muslim communities.

2 Background

In this section I will explore the history of Hindu and Muslim inter-communal violence. I will show how there is research regarding the economic and political factors for why such violence exists in the region. Furthermore, I will review literature associated with previous incidents of riots in India along with current research on the 2002 Gujarat riots.

2.1 History of Inter-Communal Violence

Since Independence in 1947, India has faced many incidences of inter-communal violence against Hindus and Muslims. In the state of Gujarat, where this paper focuses, Hindus and Muslims have rich history prior to British colonization. For Hindus, Gujarat has mythological foundations since it was the Dwarka Kingdom founded by Lord Krishna. For Muslims, many migrated from Ancient Persia and established Medieval Kingdoms that greatly contributed to trade in the Gujarat. Until British colonization, scholarship suggests that Hindu and Muslim conflict unlikely came from religious differences. However after British colonization, authorities divided the populous by emphasizing the differences between Hindus and Muslims [CITE]. From partition, the creation of India and Pakistan furthered religious tensions since it forced many Muslims away form Gujarat and into Pakistan. Spouts of violence become more common as Muslims started to become a greater minority in the region. Specifically in cities like Ahmadabad, Surat, and Vadodara, the ghettoization of Muslims furthered the disparities between Hindus and Muslims in Gujarat [CITE]. During the period of the 1970s

and 1990s, there was a constant threat of riots happening across Gujarat. [transition into the 21st century]. 2001 had the costliest earthquake recorded in Indian history hit the Western provinces of Kutch, which recorded 20,000 people [CITE]. The earthquake led many to think how such a catastrophic event could unify a divided people. However, that was not the case. In February 27th 2002, 59 people from the holy Hindu city of Ayodhya were burnt alive by a mob of people in the city of Godhra.

2.2 Literature Review

The literature on inter-communal riots between Muslims and Hindus in India focuses on the economic and political factors for why violence exists. In a broader overview on determinants of violence, a paper by Anirban Mitra and Debraj Ray explores the economic motivations of riots in India before 2002 [CITE]. The paper explains how income variation among Hindus and Muslims is a contributing factor to the intensity of riots. Specifically, they find that increased expenditures from Muslims create an environment more likely for violence to occur. The paper finds that when there is a perceived increase in wealth among Muslim households, Hindu extremists are inclined to destroy the wealth created. The implication is that in the majority of Hindu districts across time periods will respond with violence to positive economic changes among Muslim communities. This paper provides a comprehensive understanding of motivations for violence among the majority Hindu communities; however, the time series doesn't include periods after 2000, which is relevant for explaining inter-communal violence in Gujarat. Moreover, the paper looks across all regions of India, which presents problems of heterogeneity since provinces across India are vastly different and have other factors that contribute to violence. Outside of Gujarat, the province of Assam in Northeast India, language use in communities, particularly in Bengali and Assamese because of important factors on inter-communal violence occurring [CITE]. This paper highlights how language can stratify communities. Though not to the same extent, Gujarat faces linguistic stratification, with clear lines on religion. For Hindus, most

are taught in Gujarati and Hindi, while Muslims are brought up with Hindi/Urdu and learn Gujarati to exist in society. The difference being that Hindus are less likely to interact with non-Gujarati languages compared to Muslims.

Figure 1: Map of Gujarat with Muslim Monuments Destroyed

A paper looking at contributing factors to violence in the 2002 Gujarat riots is by Saumrita Jha. In his analysis, he focused on historical data from Muslim ports and leaders. His theory is that in districts and cities with a history of Muslim leaders, there is greater religious harmony in the region [CITE]. In his model, he considers factors of distance to Godhra, the BJP vote share, and other factors that contribute to violence. In his paper, he found that towns with medieval ports and that were close to Godhra had statistically significant levels of violence. Jha's paper is useful because of the use of historical data as a contributor to violence. Another paper that looks at the political theory behind the riots is "The Role of the Media During Communal Riots in India: A Study of the 1984 Sikh Riots and the 2002 Gujarat Riots" by Saifuddin Ahmend from NTU. His paper explores the theory of media, particularly regional language papers, in creating non-object media [CITE]. In the paper, Ahmend points to the Gujarat Samachar and Sandesh, the largest Gujarati language newspapers, as a perpetrator in causing violence. People with more only exposure to propagandized media are more likely to commit acts of violence compared to people with access to objective outlets. During the time, non-Hindi language papers (including Hindi,

English, Telugu, and more) were considered more objective to the events of 2002. [CITE] One report I found particularly interesting was how inter communal violence costs around 7pct of India's GDP per year. Even more striking is that in South Asia, which includes Pakistan, Afghanistan, and Sri Lank, 70pct of GDP lost from inter communal violence comes from India. Investors into South Asia will invest more in areas that are not completely susceptible to Hindu-Muslim violence.

3 Data Set

This section discusses the data involved with creating this analysis. There will be explanations on creation of data and rationale for using certain variables. There are three different types of variables used for this analysis, where Total Muslim Monuments Destroyed is Dependent, Percent Gujarat Mother Tongue and Percent Hindu minus Percent Muslims is Independent, and Good Building Stock and Casualties from 2001 Earthquake being control variables.

3.1 2001 India Census

A large portion of data comes from the 2001 Indian census. Every 10 years, India conducts one of the largest censuses in the world. Because the riots occurred in 2002, I wanted to use recent demographic and spatial data that would capture the factors within Gujarat. I am assuming that there are no significant changes between the 2001 census and population in 2002. Because the 2001 census was conducted after the 2001 earthquake, the loss of life and damages to infrastructure are taken into account within census data, particularly in the regions of Western Gujarat. I use linguistic data about mother tongue, since Gujarati as a mother tongue can indicate exposure to propaganda for people within districts. Someone show mother tongue is Gujarati is more likely to internalize inflammatory rhetoric.

3.2 Destruction of Muslim Monuments

The result of the riots in 2002 was felt mainly by the Muslim community. The riots led to the death of over 1000 people, mainly Muslims. There are unreliable accounts of deaths since many do not account how many Muslims were not considered deaths in the pogroms. Because reports on religious monuments are more reliable than deaths, there are often results that show a relationship between how Muslims were affected by the riots. I utilize data from Center of Justice and Peace (CJP), which is a non-governmental human rights organization that catalogs data from events that oftentimes the Indian government does not properly cover. For this analysis, I am using their data from 2002 about religious monuments across all regions in Gujarat [CITE]. I am combining 3 different Muslim religious monuments that signal violence against Muslim communities. I add Masjids (Mosques), Darghas (Sufi Tombs), and Madrasas (historical Islamic schools).

4 Empirical Framework

This section I show the model and explain my rationale behind choosing these independent terms. Since this model is static, there won't be significant discussion on model complexity, however, I will go over each term used.

$$Y_i = \beta_0 + \beta_1(pctGujarati)_i + \beta_2(pctHindu - pctMuslim)_i + \beta_3(goodBuildings)_i + \epsilon_i \quad (1)$$

$$Y_i = \beta_0 + \beta_1(pctGujarati)_i + \beta_2(pctHindu - pctMuslim)_i + \beta_3(casualtiesEarthquake)_i + \epsilon_i \quad (2)$$

where:

Y_i	Total number of Muslim monuments destroyed in district i
$pctGujarati_i$	Percentage of the population in district i whose mother tongue is Gujarati
$(pctHindu - pctMuslim)_i$	Percentage of Hindu population in district i minus the percentage of Muslim population
$goodBuildings_i$	Total number of buildings classified as "good" in the 2001 Census
$casualtiesEarthquake_i$	Total number of casualties from the 2001 earthquake in district i

4.1 Rationale

I used an OLS model because this model looks at the totality from an exogenous event. Using an MLR model makes sense since population does not significantly change throughout a period of 1 year. I construct my dataset using a step-wise process based on my two main control variables. I decide to do this because I want to show how different regressor change when adding more into my question. Since I focus linguistic and religious data, I include them first then add the control variables later. This ensures that the model shows variation can be control within district levels. Using a static model allows assessment of impacts from the totality of an event compared to within period of short time. Also, the 2002 riots in Gujarat are an uncommon case of violence lasting months, for many riots they are short in nature, so a long time series cannot account for such quickness.

Table 1: Stepwise Regression Results: Dependent Variable = Total Muslim Destroyed

	(1)	(2)	(3)	(4)
Intercept	14.044	36.633	37.695	-10.600
	(0.717)	(0.449)	(0.445)	(0.804)
Percent Gujarati Mother Tongue	0.080	0.096	0.105	0.521
	(0.859)	(0.833)	(0.822)	(0.204)
Percent Hindu minus Percent Muslims		-0.359	-0.369	-0.536
		(0.423)	(0.420)	(0.163)
Casualties from 2001 Earthquake			-0.0003	-0.0006
			(0.571)	(0.190)
Total Stock of "Good" Buildings				0.0001
				(0.004)**
Observations	23	22	21	23
R^2	0.002	0.034	0.051	0.402
Adjusted R^2	-0.046	-0.063	-0.099	0.269

Note: p-values in parentheses. ** $p < 0.01$

5 Results

5.1 Regression Tables

5.2 Discussion of Results

Looking at the results, there seems to be little evidence to show statistical significance between the regressor I was focusing on. Through using a step-wise model for control variables showed that Percent Gujarati and Difference in Percent of Hindu and Muslim are insignificant. However, it is important to note how the signs do not change across all regressor variables. There is a positive association with percent Gujarat as mother tongue and Muslim monuments destroyed, but there is a negative association with difference of Hindu and Muslims for Muslim monuments destroyed. This implies a LOOSE prediction that areas with high percentage of mother tongue Gujarat speakers and Muslims compared to Hindus would experience greater riot intensity, but there is nothing strong.

Looking at different tests, there seems to be problems. The QQ-plot which tests normality

Table 2: Stepwise Regression Results: Dependent Variable = Total Muslim Destroyed

	(1)	(2)	(3)	(4)
Intercept	14.044	36.633	-8.447	-10.600
	(0.717)	(0.449)	(0.846)	(0.804)
Percent Gujarati Mother Tongue	0.080	0.096	0.470	0.521
	(0.859)	(0.833)	(0.258)	(0.204)
Percent Hindu minus Percent Muslims		-0.359	-0.504	-0.536
		(0.423)	(0.195)	(0.163)
Total Stock of "Good" Buildings			0.0001	0.0001
			(0.008)**	(0.004)**
Casualties from 2001 Earthquake				-0.0006
				(0.190)
Observations	23	22	23	23
R^2	0.002	0.034	0.341	0.402
Adjusted R^2	-0.046	-0.063	0.236	0.269

Note: p-values in parentheses. ** $p < 0.01$

follows a line, which means the data looks to follow a normal distribution. However, the residuals are not set at zero because they are negative. This could be because in some regions there were no incidents of riots which could lower that linear model than in actuality. The model also incurs problems of heteroskedasticity, which would be explained by more variations than seen within the model.

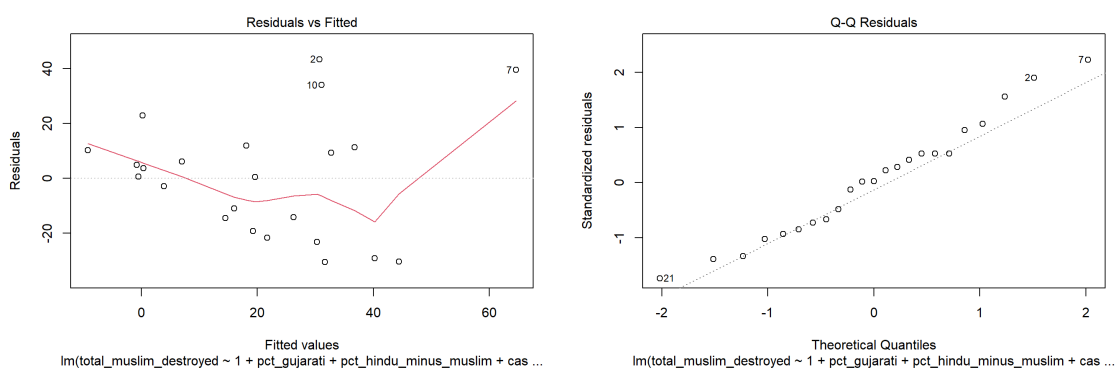

Figure 2: Residual plot and Q-Q plot for regression diagnostics

5.3 Implications

While literature on riots in South Asia focuses on language and religious composition, it is important to note that there does not seem to be statistical significance with these factors. These findings suggest that there should be more empirical work done to determine if there is a causal impact that language and religion have on riots. However, the signs of dependent variables suggest that initial theory is follows when assessing minor causes.

6 Conclusion

Though not significant, these findings are important in giving life to the thousands who lost their lives to riots from 2002. In the literature there should be more work done to solidify a relationship between theory and practice.

There were limitations with this study. For one, there was not enough data present for deaths to accurately conclude whether deaths could be predicted, using Muslim monuments presents problems in that stock of buildings influences the destruction. Other concerns could be reverse earthquake causing a buildings to be destroyed before the incident occurs. Modeling for future riots might be hard since the control variable of earthquake is specific to the time.

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